Board A plus the minimum of board B. 48 V max. XT30 bus input, TVS and cans, two LMR38010 rails, USB-C, RS-485 relay on two SH 6, expansion header.

01_bridge_cpu

Three half-bridges: 6x BSC030N08NS5, 3x EG2103, gate networks, inline shunts + INA241A3; the RP2350A, QSPI flash, crystal, 3V3 LDO

File: 01_bridge_cpu.kicad_sch

02_power

XT30 bus input, SMDJ54A, 2x 100 uF polymer, bus divider, 2x LMR38010: 12 V gate rail (GATE_OFF stops it), 5 V logic

File: 02_power.kicad_sch

03_encoder_io

MT6701 encoder; USB-C (data + 5 V), RS-485 relay on two SH 6, expansion header, RGB LED, BOOTSEL, test pads

File: 03_encoder_io.kicad_sch

CERN-OHL-P
See ../../single.html
Sheets drawn by tools/schlayout_s.py from the netlist in tools/schematic_s.py

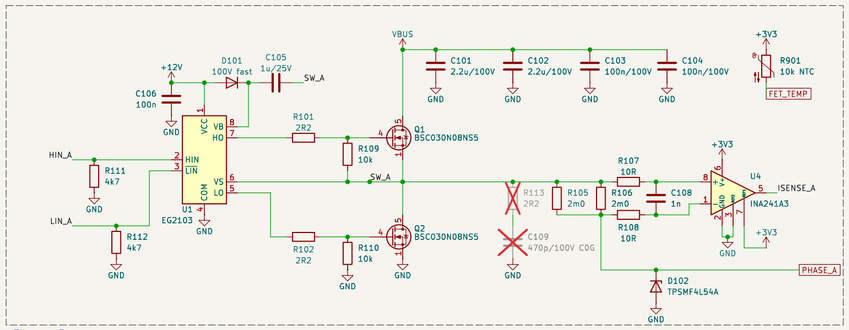
Sequoia Hope Alexander

Sheet: /
File: servodrive_S.kicad_sch

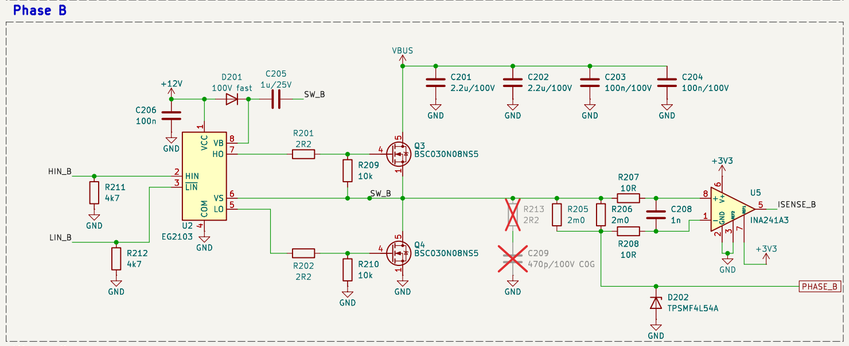
Title: servodrive — board S, single board

Size: A4	Date: 2026-09-07	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 1/4

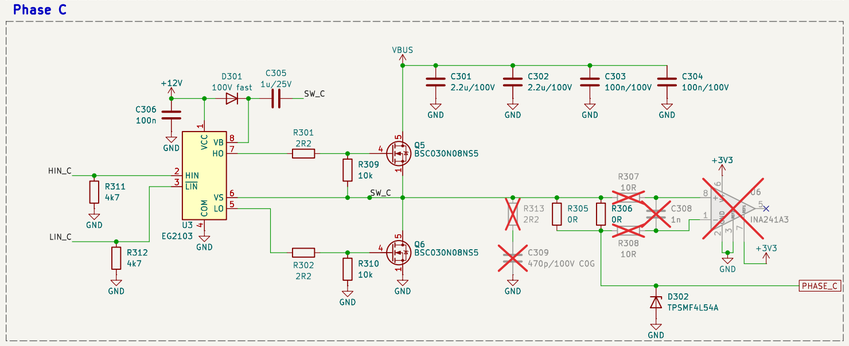
Phase A



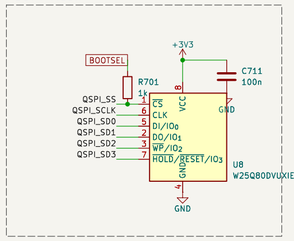
Phase B



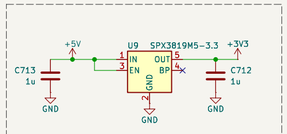
Phase C



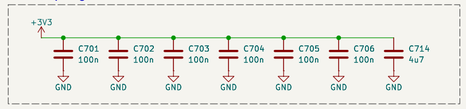
QSPI flash



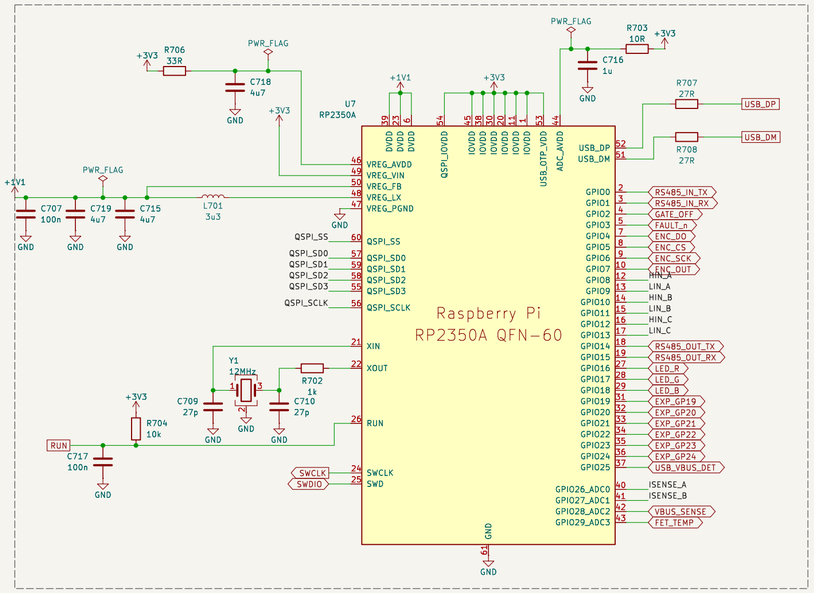
3V3 LDO



Decoupling



RP2350A



Notes: the power bridge

Each phase is an EG2103 half-bridge driver on a pair of BSC030N08NS5: bootstrap diode and capacitor (D101, C105 on phase A), 2R2 gate resistors, 10k gate-source pull-downs.

HIN and LIN are held low (R111, R112): with both low the EG2103 turns the low side on, so a CPU in reset brakes the motor.

Inline shunt: two 2m0 in parallel between the switch node and the motor lead, Kelvin-sensed through 10R, 10R and 1n into an INA241A3. REF1 on +3V3 and REF2 on GND put its output at mid-rail, so it reads current either way.

Phase C: the shunts are 0R links and U6, R307, R308, C308 are not fitted. The RP2350A has no fourth ADC channel; phase C current is −(Ia + Ib).

R113 + C109 (and B, C): RC snubber across each low side, footprints only.

R901: NTC beside the FETs, read on ADC3 as FET_TEMP.

Notes: the CPU

RP2350A as Raspberry Pi's minimal design (RP-006440): the core supply is the chip's own switching regulator, VREG_LX through L701 to +1V1, which VREG_FB senses and DVDD takes. VREG_AVDD is +3V3 through R706 and C718.

One 100n at each IOVDD pin (C701–C706): C714 4u7 at VREG_VIN.

ADC_AVDD is +3V3 filtered by R703 and C716 (+3V3A).

BOOTSEL: SW1 on sheet 03 pulls QSPI_SS low through R701 at reset.

The encoder: ENC_DO on GPIO4, ENC_SCK on GPIO6, ENC_CS on GPIO5, ENC_OUT on GPIO7 — SPIO's RX and SCK, so SPIO reads it.

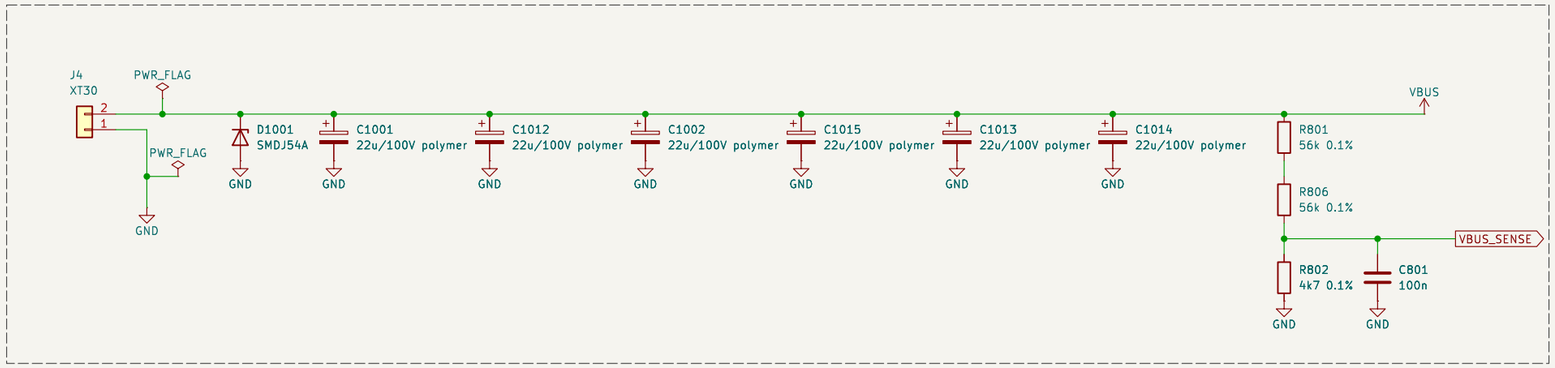
GPIO19–24 go to the expansion header (sheet 03).

CERN-OHL-P
Drawn by tools/schlayout_s.py from the netlist in tools/schematic_s.py
Sequoia Hope Alexander
Sheet: /01_bridge_cpu/
File: 01_bridge_cpu.kicad_sch

Title: servodrive — board S, single board — 01_bridge_cpu

Size: A2	Date: 2026-09-25	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 2/4

Bus input



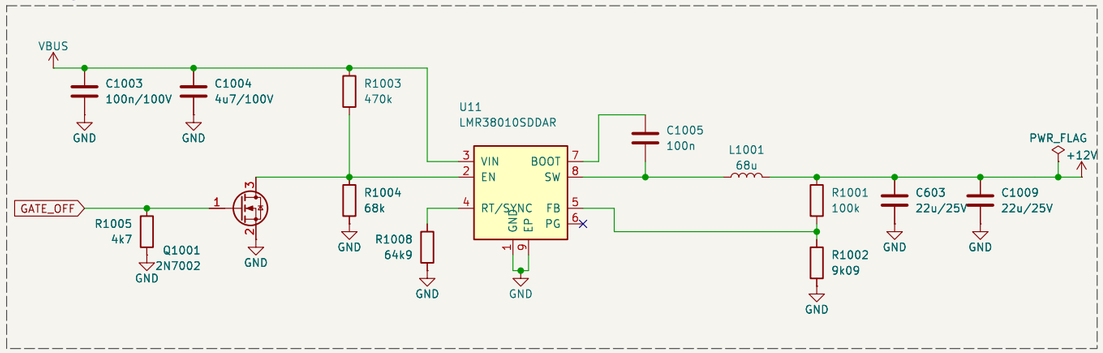
Notes

Bus: 48 V operating on 80 V FETs. D1001 (SMDJ54A, 54 V standoff) clamps it; C1001, C1012, C1002, C1015, C1013 and C1014 are the bulk. VBUS_SENSE = VBUS x 4k7 / 116k7 (0.1 %), on ADC2.

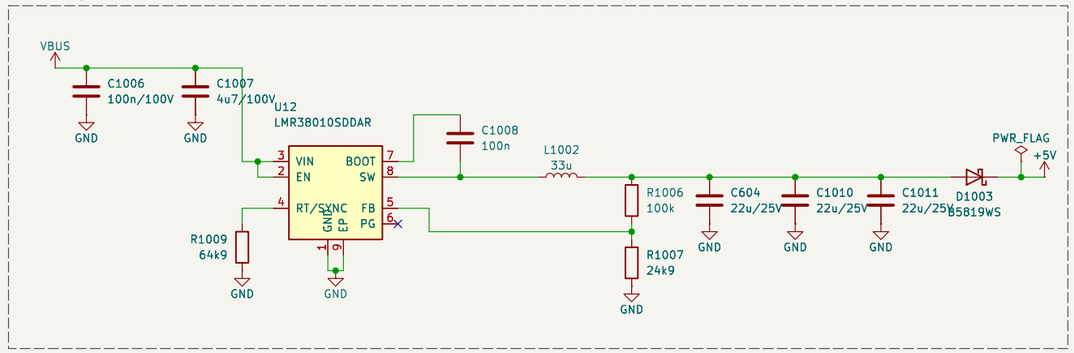
12 V gate rail: LMR38010, starting at 9.9 V on the R1003/R1004 divider. GATE_OFF (GPIO2) high turns Q1001 on and stops the rail, taking the gate drivers' supply away. R1005 holds it low through reset, so the rail comes up and the drivers brake the motor.

5 V logic rail: LMR38010, diode-ORed with USB (D1002, sheet 03) through D1003. The 3V3 LDO on sheet 01 runs from +5V.

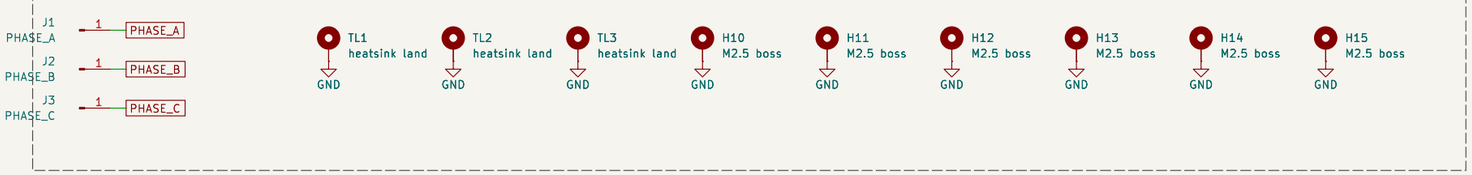
12 V gate rail



5 V logic rail



Mounting and motor leads



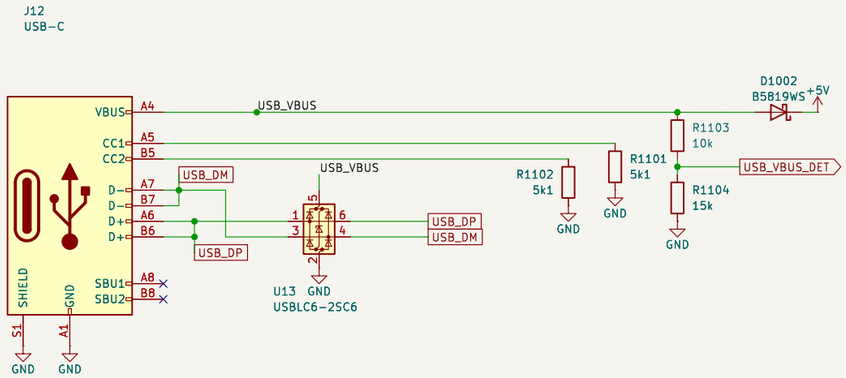
CERN-OHL-P
Drawn by tools/schlayout_s.py from the netlist in tools/schematic_s.py
Sequoia Hope Alexander

Sheet: /02_power/
File: 02_power.kicad_sch

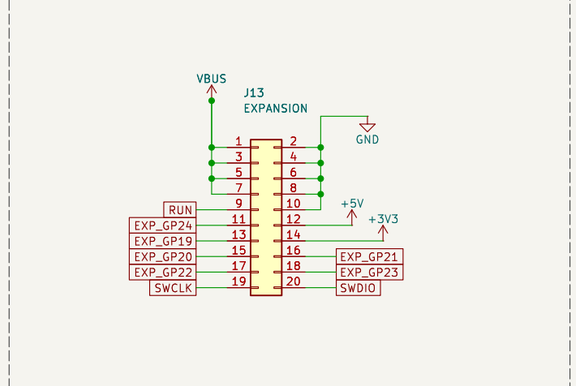
Title: servodrive — board S, single board — 02_power

Size: A3	Date: 2026-09-25	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 3/4

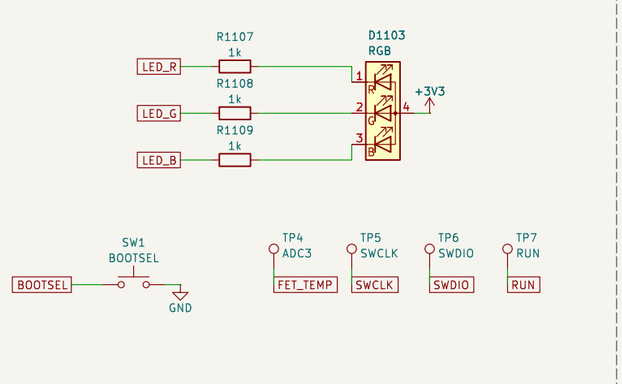
USB-C: data and 5 V logic, no PD



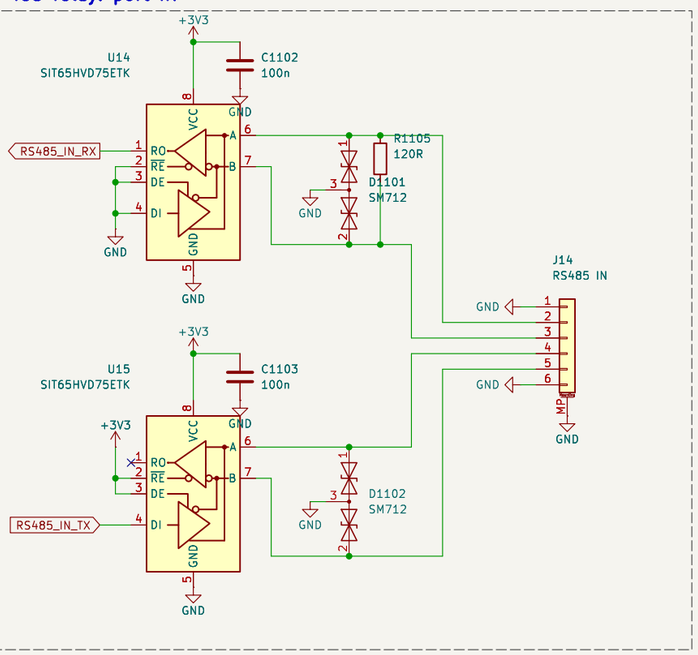
Expansion header: PD or Ethernet board



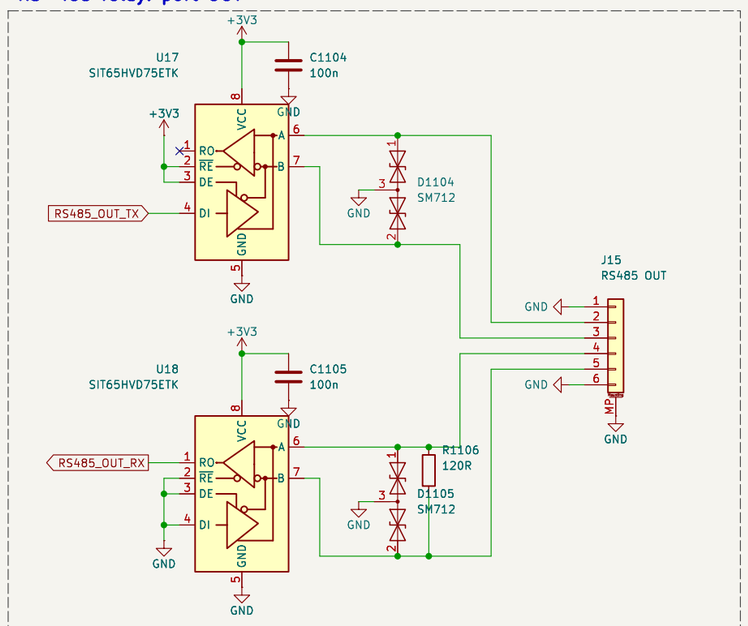
Status LED, BOOTSEL, test pads



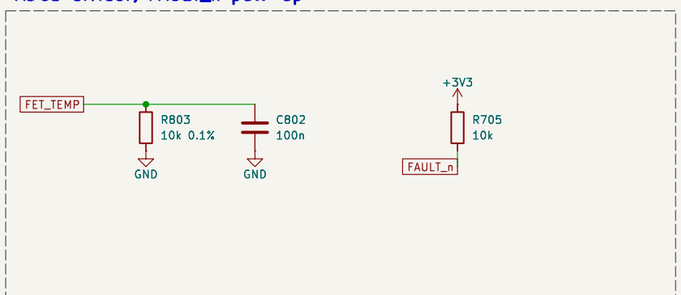
RS-485 relay: port IN



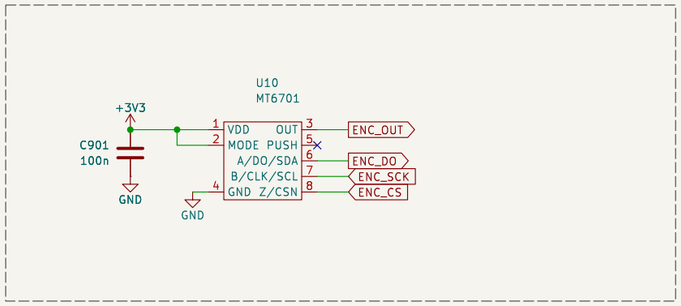
RS-485 relay: port OUT



ADC3 divider, FAULT_n pull-up



Encoder: MT6701 on the shaft axis



RS-485 is a CPU relay: each port is two point-to-point pairs, one in and one out, so a drive passes the chain on in software. Drivers are always enabled; each receiver has its 120R termination (R1105, R1106).

The MT6701 sits on the motor-facing side, centred on the shaft over a diametric magnet, MODE tied to VDD; the CPU reads it over SSI (sheet 01).

R803 with the NTC (R901, sheet 01) makes FET_TEMP.

CERN-OHL-P
Drawn by tools/schlayout_s.py from the netlist in tools/schematic_s.py
Sequoia Hope Alexander
Sheet: /03_encoder_io/
File: 03_encoder_io.kicad_sch

Title: servodrive — board S, single board — 03_encoder_io

Size: A3	Date: 2026-09-25	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 4/4